

WebApi Handson Solution

4. WebApi Hands-on Solution

Objectives Addressed

- * Action method to perform CRUD operations (PUT).
- * Extract data to custom model using `[FromBody]`.
- * Validate ID and return appropriate HTTP status codes like `BadRequest`.

Solution Implementation

The objective here is to update employee data based on user input using HTTP PUT.

1. Employee Controller (CRUD Operations)

```
```csharp
using Microsoft.AspNetCore.Mvc;
using System.Collections.Generic;
using System.Linq;
using WebApi_Handson.Models;

namespace WebApi_Handson.Controllers
{
 [Route("api/[controller]")]
 [ApiController]
 public class EmployeeController : ControllerBase
 {
 // Simulated Database
 private static List<Employee> employees = new List<Employee>
 {
 new Employee { Id = 1, Name = "John", Salary = 50000, Permanent = true },
 new Employee { Id = 2, Name = "Smith", Salary = 60000, Permanent = false }
 };

 [HttpGet]
 public ActionResult<IEnumerable<Employee>> Get()
```

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```
{
 return employees;
}

// HTTP PUT to update an employee
[HttpPut("{id}")]
public ActionResult<Employee> Put(int id, [FromBody] Employee employeeData)
{
 if (id <= 0)
 {
 return BadRequest("Invalid employee id");
 }

 var existingEmployee = employees.FirstOrDefault(e => e.Id == id);

 if (existingEmployee == null)
 {
 return BadRequest("Invalid employee id");
 }

 // Update Employee Data
 existingEmployee.Name = employeeData.Name;
 existingEmployee.Salary = employeeData.Salary;
 existingEmployee.Permanent = employeeData.Permanent;
 existingEmployee.DateOfBirth = employeeData.DateOfBirth;
 // Additional mappings if needed...

 return Ok(existingEmployee);
}
}
```

## Testing using Swagger / Postman

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1. Run the application.
2. Open **Swagger** or **Postman**.
3. Execute a `PUT` request to `/api/employee/1` with the following JSON body:

```
```json
{
  "name": "John Updated",
  "salary": 75000,
  "permanent": true
}
```
```

4. **Expected Outcome**: The request returns `200 OK` along with the updated employee JSON object.
5. If you use an ID like `-1` or `999`, the server will return a `400 Bad Request` with the message `"Invalid employee id"`.